Room Watchdog System

System Design Report

Ella Lewis

University of Waterloo – Electrical Engineering

June 2026

Abstract

This project incorporates a mmWave sensor, a Passive Infrared (PIR) sensor, and an ESP32-S3-WROOM microcontroller to create a battery-powered room occupancy monitoring system. The objective is to detect entry into a room and notify the user remotely through Wi-Fi. The report documents the system requirements, hardware design, power architecture, PCB layout considerations, design challenges, validation procedures, and future improvements.

Contents

Chapter 1

Project Description

This project incorporates a mmWave sensor and an Infrared sensor along with an ESP32-S3-WROOM for the main unit.

Goal

This circuit should be able to monitor the presence and motion at a bedroom door because snacks were suspected to be stolen during the day while the user was away.

The final product should be able to detect a *walk-in* event to the room. The microcontroller should send a signal by Wi-Fi which will then be transmitted to a phone so that the user is notified.

The product should be easy to use, inexpensive to manufacture, and easy to maintain.

A major challenge in this design is that the board is Li-Ion powered, therefore the battery management circuit required significant effort in device selection and power architecture.

The board consists of two layers due to its relative simplicity. The main layout concerns are related to:

- ESP32 antenna keep-out region
- mmWave sensor clearance area
- PIR sensor clearance area
- Power integrity across the board

The logic circuitry operates at 3.3 V while the supply can provide 5 V. Heat generation is minimal because the system operates at relatively low voltage and current levels.

Chapter 2

Power System Design

2.1 Design Changes

After further research, the design was modified so that the board is powered by the Li-Ion battery when USB-C is disconnected and the battery is charged whenever USB-C is connected. This ensures continuous operation even when external power is removed.

A challenge in the original design was that the Li-Ion charger IC only provides the battery voltage:

Battery state	Voltage
Fully charged	4.2 V
Nominal	3.7 V
Discharged	~3.0 V

Table 2.1: Li-Ion battery voltage range

The AP2112 LDO accepts an input range of approximately 2.5 V to 6 V, therefore the battery voltage is sufficient to generate the required 3.3 V rail for the ESP32 and supporting logic.

2.2 LD2410 Power Requirement

A critical issue discovered during debugging was that the LD2410 mmWave sensor requires a regulated 5 V supply. The battery voltage never exceeds 4.2 V, so the sensor cannot be powered directly from the battery.

A temporary workaround is to provide an external 5 V supply during debugging. For a future revision, a boost converter should be added so that a regulated 5 V rail can be generated from the Li-Ion battery.

Chapter 3

PCB Layout Considerations

Power integrity is maintained by placing decoupling capacitors close to the power pins of the ESP32 and sensors. Ground copper pours are used to provide a low-impedance return path.

3.1 Signal Integrity

The USB differential pair is routed together with closely matched lengths because it carries high-speed USB signals.

3.2 Mechanical Considerations

Both sensors are mounted facing the same direction to simplify installation and improve usability.

3.3 Layout Improvements

Possible future improvements include reducing the power loop area to minimize loop inductance, although this is not a significant issue in the current low-power design.

Chapter 4

Sensor Selection

The design uses both a PIR sensor and a mmWave sensor because each technology offers complementary advantages.

4.1 PIR Sensor

- Detects changes in infrared radiation
- Very low power consumption
- Suitable for continuous monitoring

4.2 mmWave Sensor

- Detects very small motion
- Higher sensitivity than PIR
- Useful for detecting subtle movement

Using both sensors together improves occupancy detection reliability while reducing false detections.

When no significant motion is detected by the mmWave sensor, the ESP32 can remain in a low-power sleep state while continuing to monitor the PIR sensor.

Chapter 5

ESP32-S3-WROOM Selection

The ESP32-S3-WROOM was selected because it integrates Wi-Fi communication and adequate processing capability for sensor monitoring and notification tasks. Using an integrated Wi-Fi microcontroller reduces component count, PCB area, and overall system cost.

The ESP32-S3 also provides low-power sleep modes, which are beneficial for battery-operated applications.

5.1 Boot and Reset Functions

- **BOOT button**: selects normal boot or firmware download mode.
- **RESET button**: resets the microcontroller and is used together with BOOT during firmware flashing.

Chapter 6

USB-C Configuration

This board operates as a USB-C *sink*. The CC pins are connected through pull-down resistors to indicate to the host that the device requests power.

If the board were a USB-C *source*, the CC pins would instead use pull-up resistors.

Chapter 7

Trace Width Selection

Signal / Rail	Width	Notes
LD2410 5 V supply	0.2 mm	∼80 mA load
VBAT and +5 V rails	0.5 mm	Higher current path

Table 7.1: Selected PCB trace widths

Chapter 8

Testing and Validation

8.1 Visual Inspection

- Solder joints
- Component orientation
- Sensor mounting direction
- Connector placement

8.2 Electrical Verification

- Continuity check
- Input current verification (<100 mA)
- LDO input/output voltage measurement
- MCU supply verification

8.3 USB-C Power Test

Connect USB-C from a computer and verify that the power LED illuminates and that all power rails are present.

8.4 Firmware Flashing Procedure

1. Hold BOOT.
2. Press RESET.
3. Release RESET.

4. Release BOOT.
5. Flash firmware.
6. Press RESET to execute the new firmware.

8.5 Functional Test

Verify that the sensors detect occupancy and that notifications are transmitted through Wi-Fi.

Chapter 9

PCB Figures

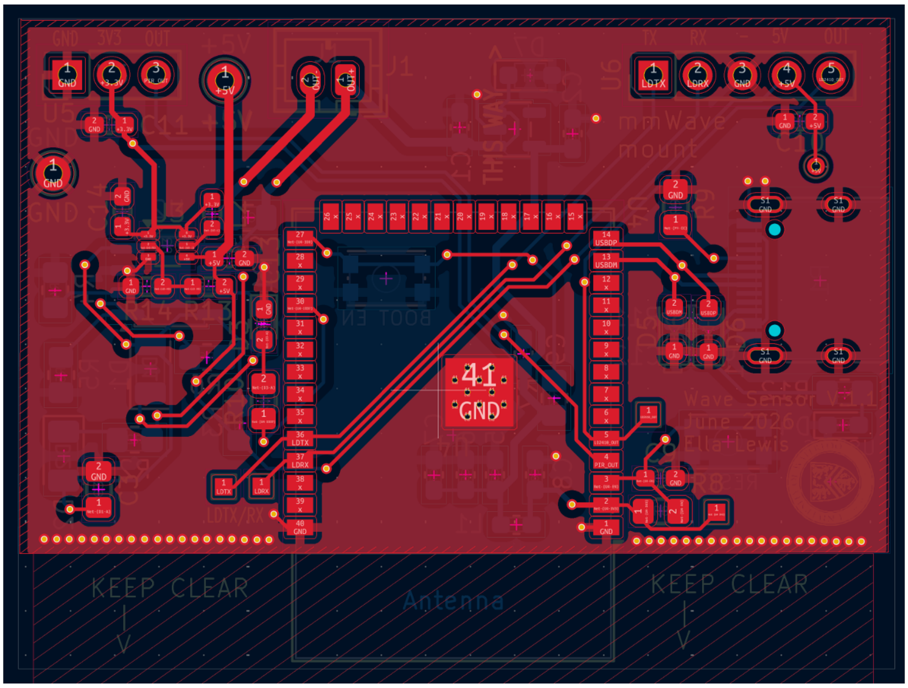

Figure 9.1: Front 2D Layout

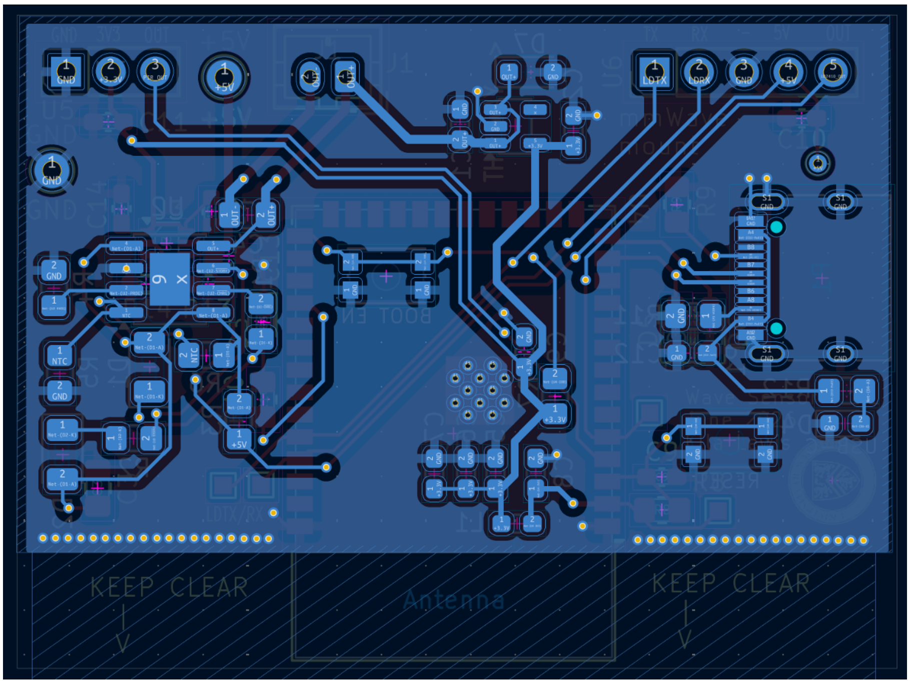

Figure 9.2: Back 2D Layout

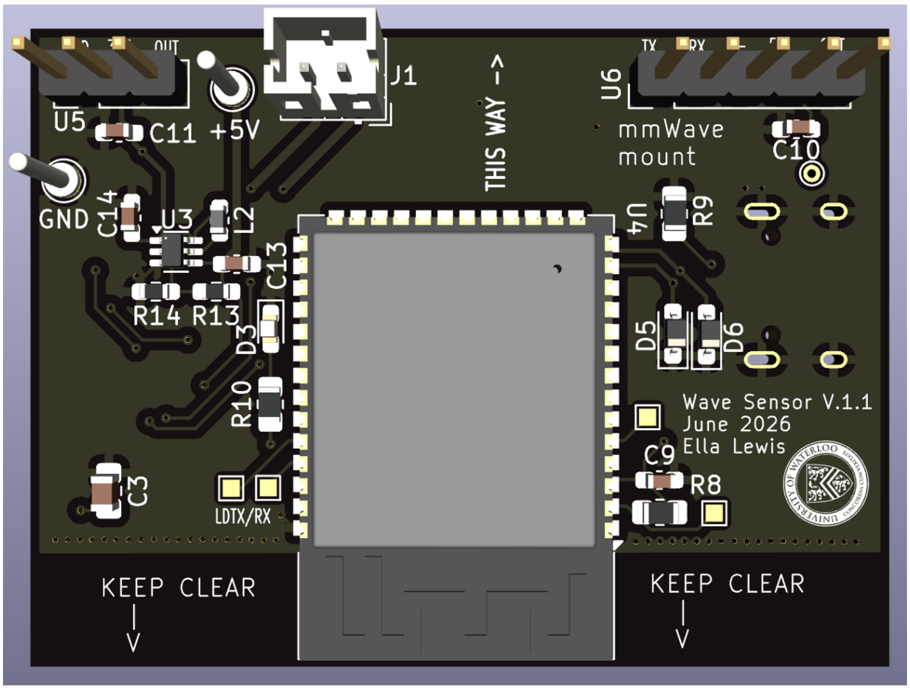

Figure 9.3: Front 3D View

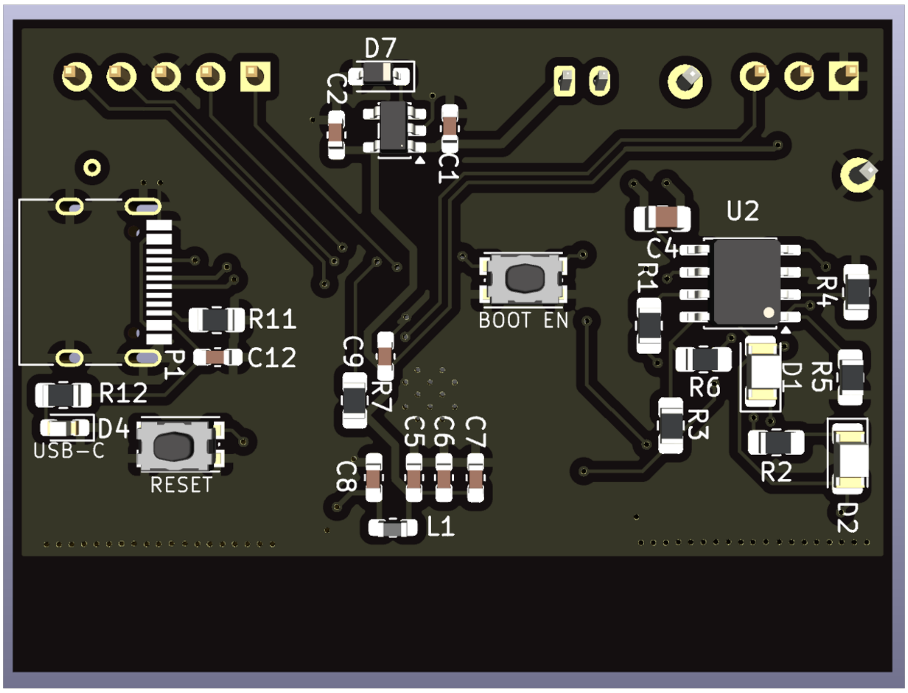

Figure 9.4: Back 3D View

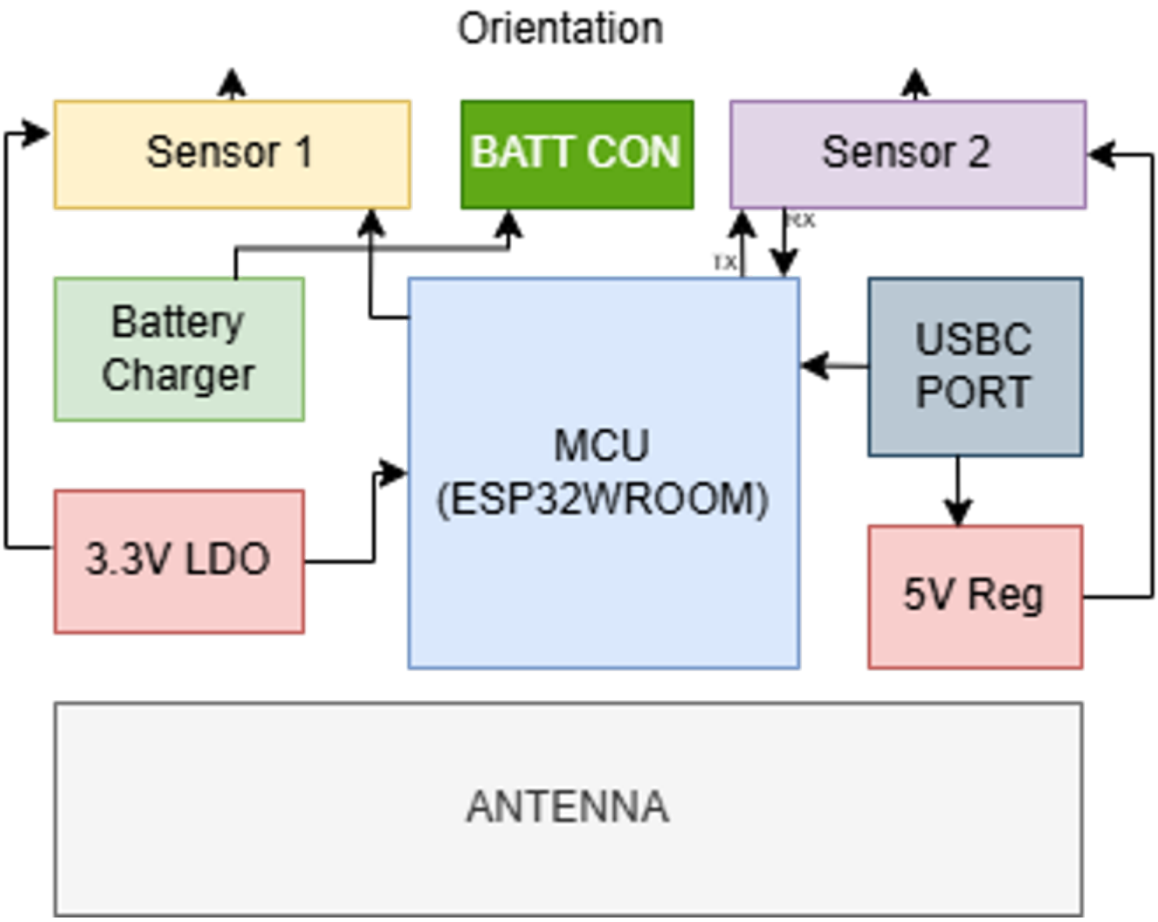

Figure 9.5: System Overview

Chapter 10

Future Improvements

- Add an integrated 5 V boost converter
- Reduce power loop area
- Optimize sensor placement
- Improve low-power firmware operation
- Add a battery fuel gauge
- Design an enclosure

Chapter 11

Revision History

Revision	Description
1.1.1	Added voltage regulator

Table 11.1: Revision history

End of Report